SANDHYA KAPSE

Data & AI Engineer · Azure Fabric · Databricks · Snowflake · PySpark · SQL · RAG · LLMs · Responsible AI

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~4 Years	86%	50M+	35%
Data & AI Engineering	Propensity Model Accuracy	Daily Events Processed	Query Runtime Reduced

PROFESSIONAL SUMMARY

Data & AI Engineer with **nearly 4 years** building and operating production data platforms on Azure. At **Tredence**, architected AutoFX — a Medallion pipeline for a global bank processing **50M+ daily payment events**, a six-checkpoint ML pipeline (86% accuracy), RAG-grounded agents on Azure AI Foundry and SHAP-based Responsible AI dashboards. At **Microsoft**, owned ADF ingestion for enterprise customers and root-caused multi-tenant failures that led to **platform fixes shipped globally**. Unusual mix: I build pipelines *and* know how they fail in production. Pursuing **DP-700**.

CORE TECHNICAL SKILLS

Azure Data & AI:	Microsoft Fabric (OneLake · Pipelines · Eventstream) · Azure Data Factory · Databricks · Synapse · ADLS Gen2 · Azure AI Foundry · Azure OpenAI · Azure AI Search
AI / ML / GenAI:	LLM Integration · RAG / Grounding · Prompt Engineering · NLP · SHAP Explainability · Feature Engineering · LangChain / LangGraph (Familiar) · Responsible AI (Fairness · Interpretability · Governance)
Languages & Orchestration:	Python · PySpark · SQL · Spark SQL · Apache Airflow · Databricks Jobs · FastAPI · MCP Endpoints · Kafka (Familiar) · SSIS (Familiar)
Data Platforms & Modelling:	Snowflake · Delta Lake · Fabric Lakehouse · Medallion Architecture · Dimensional Modelling · CDC · SCD Type 2 · REST APIs · SAP Extraction · MuleSoft / Salesforce APIs
Governance & DevOps:	Data Governance · RBAC · Data Lineage · Audit Telemetry · Schema Validation · DataOps · CI/CD (Azure DevOps) · Observability · RCA · Runbooks · Agile

PROFESSIONAL EXPERIENCE

Data & AI Engineer — Jul 2026 – Present

Tredence · Bangalore

- Architected **AutoFX** — AI-powered payment data platform for a global bank: Medallion pipelines (staging → canonical → curated) processing **50M+ daily FX events** on Azure Fabric OneLake and ADLS Gen2.
- Designed a **six-checkpoint ML pipeline** with Azure AI Foundry LLMs, a feature store and a propensity model (**86% accuracy**) enabling straight-through FX conversion.
- Built **RAG-grounded agentic workflows** via MCP endpoints and FastAPI, returning **SHAP-grounded, Responsible AI-aligned** KPI insights with fairness and interpretability governance.
- Delivered curated semantic models across six feature domains, plus RBAC, lineage, audit telemetry and **CI/CD** governance on Azure DevOps across all pipeline layers.

Data Engineer — Mar 2025 – Jun 2026

Microsoft (via Aptly Professional Technology) · Bangalore · 1 yr 4 months

- Owned **Azure Data Factory ingestion** for global enterprise customers — designed and configured **10+ production connectors** (SAP ERP, REST APIs, SFTP, Fabric Lakehouse) and tuned them for reliability at scale.
- Root-caused complex **multi-tenant pipeline failures** through telemetry and log analysis; findings drove **platform fixes shipped globally** to enterprise customers.
- Led incident triage and **RCA** on live production data issues; wrote runbooks and SOPs, and partnered with architects and engineering to mitigate risk and prevent repeat failures.

Data Engineer — Sep 2024 – Mar 2025

CoffeeBeans Consulting · Bangalore · 7 months

- Built Python + Snowflake ETL with validation, **source-to-target reconciliation** and completeness checks; cut query runtime **35%** on 5M+ daily eCommerce events.
- Built **MuleSoft** integrations with Salesforce (Bulk and REST APIs) including error handling and retry logic; added ML-ready feature layers with audit logging and quality governance.

Data Engineer — Jan 2022 – Jun 2023

ITC Infotech · Bangalore · 1.5 years

- Led **cloud migration** (SQL Server / PostgreSQL → Snowflake / Azure Data Lake): profiling, schema-conflict resolution, **SCD Type 2** and star-schema modelling.
- Built **Airflow-orchestrated ADF pipelines** with CI/CD on Azure DevOps — SoX-aware audit logging, lineage and zero-regression releases across dev, UAT and production.

KEY PROJECTS · EDUCATION · CERTIFICATIONS

AutoFX Intelligent Payment Platform · Tredence

Fabric OneLake → 6-checkpoint ML pipeline → RAG agents (Azure AI Foundry) → SHAP Responsible AI dashboard. 50M+ payment events/day.

Cross-Cloud Streaming Pipeline · AWS

Twitter API → AWS Lambda → S3 → Elasticsearch, event-driven.
github.com/sandhya71/Twitter_Data_Pipeline

Olympic Data Transformation · Azure

ADF → ADLS Gen2 → Databricks PySpark → curated analytics layer.
github.com/sandhya71/Olympic_Transformation

B.E. Computer Science & Engineering

Alva's Institute of Engineering & Technology, Mangalore
2021 · CGPA 7.1 / 10

Certifications

In preparation: DP-700 Fabric Data Engineer Associate
Targeting: AI-102 Azure AI Engineer Associate